Academic case study | Sophia Learning — Introduction to Networking
GreenLeaf Network Security Assessment

Name: Sanja Filipova

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1. Network Security Architecture

1a. Security Zones Diagram Summary

GreenLeaf's network is segmented into four security zones. The Outside zone is the untrusted internet and only allows limited inbound traffic (such as VPN and monitoring), while the Inside zone contains trusted networks for users, VoIP, servers, and corporate wireless devices and can access necessary internet services such as web and DNS. The VPN zone provides an encrypted site-to-site connection between the Portland and Eugene locations, allowing trusted traffic between the two networks. The Guest zone is isolated from internal resources and is only allowed internet access for web browsing and DNS, with all other access to the inside network blocked by the firewall.

1b. Wireless Standards and Radio Settings

Finding ID	Severity	Description	Recommendation
H-001	High	SNMPv2c is being used with community strings that are sent in clear text, which could allow an attacker to intercept them and access network devices.	Migrate to SNMPv3 with authentication and encryption and restrict SNMP access to authorized management systems.
H-002	High	The Eugene router is running an outdated IOS version with known vulnerabilities that could allow remote code execution and compromise the device.	Upgrade the Eugene router to IOS-XE 17.03.06 or later after backing up and testing the configuration.
M-008	Medium	GreenLeaf does not currently have an IDS or IPS, which means	Deploy an IDS/IPS and configure alerts for common attacks and suspicious network activity.

		network attacks may not be detected or blocked.	

2. Firewall and Access Control

2a. Firewall Rule Summary

Rule #	Source → Destination	Service	Action	Business Need
INSIDE→OUTSIDE #1	10.10.10.0/24 → Any	HTTP/HTTPS	Permit	Allows employees to access websites and other web-based business services.
INSIDE→OUTSIDE #2	10.10.10.0/24 → Any	DNS	Permit	Allows user devices to resolve domain names.
INSIDE→OUTSIDE #3	10.10.30.0/25 → Any	HTTPS	Permit	Allows GreenLeaf servers to securely download updates.
OUTSIDE→INSIDE #1	198.51.100.10 → 203.0.113.10	ISAKMP/ESP	Permit	Allows the Eugene site to establish the encrypted VPN connection to Portland.
VPN→INSIDE #1	10.20.0.0/16 → 10.10.30.0/25	ERP/SQL	Permit	Allows Eugene users and systems to access required Portland ERP/database services.

2b. Issues and Weaknesses

GreenLeaf has an overly permissive site-to-site rule that allows any traffic from the Portland 10.10.0.0/16 network to the Eugene 10.20.0.0/16 network, which does not follow the principle of least privilege. The firewall analysis also notes that users are allowed to access any website through HTTP/HTTPS because no web content filtering or category blocking is implemented. GreenLeaf should restrict inter-site traffic to required services and consider adding web filtering, application-layer inspection, and additional monitoring.

3. Remote Access & Encryption

3a. VPN Configuration Summary

GreenLeaf uses an IKEv2 site-to-site VPN to securely connect the Portland and Eugene locations. Phase 1 uses AES-256-CBC encryption, SHA-256 integrity, DH Group 14, and a lifetime of 86,400 seconds, while Phase 2 uses ESP-AES-256 with ESP-SHA256-HMAC in tunnel mode and includes Perfect Forward Secrecy using DH Group 14. The VPN is authenticated with a pre-shared key (PSK), with Portland at 203.0.113.10 and Eugene at 198.51.100.10 as the tunnel endpoints. The tunnel protects traffic between Portland's 10.10.0.0/16 network and Eugene's 10.20.0.0/16 network.

3b. Encryption Protocols in Use

Service / Traffic	Protocol	Encryption
Site-to-Site VPN	IPsec / IKEv2	AES-256-CBC (256-bit)
Web Applications (ERP)	TLS 1.2 / 1.3	256-bit
SSH Management	SSH	AES-256 (256-bit)
SMB File Shares	SMB 3.0	128-bit encryption

4. Authentication & Device Hardening

4a. Authentication Flow (Wireless 802.1X)

GreenLeaf uses 802.1X authentication to control access to its corporate and warehouse wireless networks. The wireless device acts as the supplicant, the access point acts as the authenticator, and the RADIUS server verifies the user's credentials through Active Directory. PEAP creates a secure TLS tunnel, and MSCHAPv2 is then used to authenticate the user's username and password inside that tunnel. After successful authentication, RADIUS sends an Access-Accept and assigns the appropriate VLAN, such as VLAN 40 for corporate users and IT staff or VLAN 50 for warehouse devices.

4b. Hardening Strengths

- SSH version 2 is enabled for secure remote management of network devices.
- Passwords and enable secrets are encrypted, including Type 9 (scrypt) encryption for the enable secret.
- Centralized logging is enabled, with network devices sending logs to GreenLeaf's syslog server.

4c. Hardening Gaps

- SNMPv2c is still enabled instead of the more secure SNMPv3, which leaves management traffic less protected.
- Telnet is still enabled on some access switches and router VTY configurations, which is a security risk because Telnet sends management traffic and credentials without encryption.
- DHCP Snooping is not enabled on the switches, leaving the network more vulnerable to rogue DHCP servers and DHCP-based attacks.